Leak-Free Fine-Tuning of a Phase-Aware Enhancer and Diagnosing Test-Set Mismatch for Indic Noise-Event Removal

Avinash Kumar Sharma (Team Vedanga)
Wadhwani School of Data Science and AI, IIT Madras, Chennai, India
avics2020@gmail.com

***Abstract*—We describe our solution to Track 2 (Noise Event Removal) of Datathon@IndoML 2026, scored as SI-SDR on synthetic clips plus $100 \times \Delta$WER of the mandated SraVaani-1.0 ASR. We fine-tune the phase-aware MP-SENet (2.26 M parameters), initialised from its DNS-Challenge checkpoint, in four warm-started stages on simulated Vaani mixtures built with a leak-free protocol: validation noise events are clustered and only training clusters are used, because 98.8% of the validation events are reused across clips. Inference uses 8-s windows and shift test-time augmentation (TTA). The selected submission scores 15.21 (SI-SDR 13.18 dB, ΔWER $+2.02\%$). We also diagnose why local and leaderboard scores differ by about 5 dB: test clips are three times longer, mostly event-free, and their references keep native Vaani noise. A local set built from noisy Vaani clips plus unseen library events reproduces the leaderboard SI-SDR of our model (12.66 vs. 13.18 dB), and an "event-only" training recipe improves it by 0.28 dB.**

I. Introduction

Track 2 asks for 16-kHz enhanced audio that suppresses noise events (horns, animals, appliances, alarms, etc.) in spontaneous Indic speech while preserving the speech. Submissions are scored as Combined $=$ SI-SDR$_{\text{syn}} + 100\,\Delta$WER, where SI-SDR [8] is averaged over the 1,099 synthetic test clips and ΔWER is the pooled WER reduction of SraVaani-1.0 over all 3,178 test clips (1,099 synthetic, 2,079 natural); transcripts are audited by re-running the ASR on the submitted WAVs. The test set provides audio only (no timestamps). Our solution has three parts: a leak-free validation protocol, staged fine-tuning of a phase-aware enhancer on Vaani-like simulations, and an inference recipe tuned for long clips. We also report what failed and a diagnosis of the local/leaderboard gap.

II. Training Data and Preprocessing

Vaani. Clean speech comes from the Gold split of ARTPARK-IISc/Vaani-Noise-Event-Dataset [3]: event-free spans ($\geq$2.5 s) carved with the organisers' baseline [5] from 6,000 clips (3,968 spans, early stages) and 11,000 clips (8,733 spans, final stages).

Validation and leak-free split. The validation set has 695 synthetic pairs (noisy, clean reference, timestamps) and 1,806 natural clips. Noise events are isolated as noisy$-$clean inside the timestamps (1,819 events). Near-duplicates are clustered (cosine > 0.95 on 48×48 log-mel fingerprints; 139 clusters) and 62 clusters are held out; the 228 synthetic clips whose events all lie in held-out clusters form the evaluation set, and only the 644 events of training clusters enter the noise bank. A first model fine-tuned on a random validation split scored 12.3 dB locally but 0.83 on the leaderboard, which exposed the reuse.

Additional data. ESC-50 [6] (1,998 trimmed event segments, $\leq$2.5 s; CC BY-NC 3.0) and the MUSAN noise subset [7] (930 files $\rightarrow$ 2,456 random crops of 0.5–2.5 s; CC BY 4.0).

Augmentation. One augmented copy per noise segment: speed 0.9/1.0/1.1; with p=0.6 a 4th-order low-pass (2–7 kHz) or 2nd-order high-pass (100–500 Hz); with p=0.5 a synthetic exponentially decaying reverb (RT 0.15–0.7 s). Final pool: 10,192 segments.

Mixing. 5-s mixtures with 1–4 events of 0.3–2.5 s at event SNR $\in [-20, 5]$ dB relative to speech (mean input SI-SDR ≈ -5 dB), matching the validation statistics. Datasets: 8k (stage 1), 16k (stage 2) and 40k mixtures (stages 3–4).

III. Proposed Method

Architecture. MP-SENet [1] predicts magnitude and phase in parallel from the compressed STFT (n_{fft}=400, hop 100, compression 0.3) with four time–frequency Transformer blocks. We keep the official configuration and initialise from the DNS-Challenge [2] checkpoint `g_best_dns`. Replacing the baseline magnitude-mask BiGRU (noisy phase) by this phase-aware model was the largest single gain (3.40 $\rightarrow$ 10.00 zero-shot).

Loss. Generator terms of MP-SENet without the metric discriminator: $\mathcal{L} = 0.9\,\mathcal{L}_{\text{mag}} + 0.3(\mathcal{L}_{\text{IP}} + \mathcal{L}_{\text{GD}} + \mathcal{L}_{\text{IAF}}) + 0.2\,\mathcal{L}_{\text{com}} + 0.2\,\mathcal{L}_{\text{con}} + 0.2\,\mathcal{L}_{\text{time}}$ (MSE on compressed magnitude and complex spectra, anti-wrapping phase losses, STFT consistency, waveform L1). An ASR-feature loss on SraVaani encoder features helped the small baseline (+0.8) but on MP-SENet traded SI-SDR for ASR roughly one-to-one (λ=0.5/1.5) and was not used.

Training. AdamW [10] (β=0.8/0.99), OneCycle [11] (5% warm-up), gradient clipping 5, inputs RMS-normalised. Stage 1 (E8): DNS init, LR 2×10^{-4}, 5 epochs, 1-s crops, batch 2, fp32 on a T4 (fp16 AMP silently skipped updates). Stage 2 (E9-B): LR 10^{-4}, 3 epochs, 16k mixtures. Stage 3: 40k mixtures, 3-s crops, batch 4, bf16 forward on an A100; the 8-epoch schedule did not decay within the time budget (held-out 17.00 dB). Stage 4 (annealing): one epoch (9,500 steps)

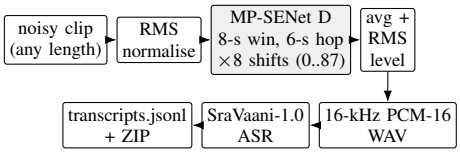


Fig. 1. Inference pipeline of the selected submission (every clip is processed by the model).

TABLE I
TRAINING STAGES OF THE SUBMITTED MODEL D (ALL WARM-STARTED; T4 FP32, A100 BF16).

Stage	Init	Mix	Crop×bs	LR	Held-out
E8 (T4)	DNS	8k	1 s×2	2×10^{-4}	16.68
E9-B (T4)	E8	16k+aug	1 s×2	10^{-4}	16.93
E10-R (A100)	E9-B	40k	3 s×4	10^{-4}	17.00
Anneal→D	E10-R	40k	3 s×4	$5\times10^{-5}\rightarrow0$	17.26

with LR 5×10^{-5} decaying to ≈ 0 gives model D (17.26 dB). Lesson: size the schedule to the measured throughput.

Reproducibility. Training and inference code, configuration files, the final checkpoint and the original notebooks are released in our public repository; the README lists the software and hardware (A100-SXM4 80 GB with PyTorch 2.8/CUDA 12.8 for the final stages and inference; Kaggle T4 for early stages).

Inference. Each clip is RMS-normalised and enhanced on 8-s windows (6-s hop, linear cross-fades); 60% of clips exceed 4 s, and 8-s windows beat 4-s ones. Shift TTA averages the realigned outputs of 8 copies delayed by 0–87 samples (sub-hop STFT alignments), which reduces artefacts. Outputs are rescaled to the input RMS. Transcripts come from SraVaani-1.0 [4]. Every clip is processed by the model.

Novelty: test-set diagnosis and event-only training. Running D on 800 test clips (analysis only) showed median length 11.9 s (validation 4.4 s) and loud-noise occupancy of 10% of frames (validation synthetic 39%), while keeping the input where D removes little *lowered* the test SI-SDR. A local set of *noisy* Vaani clips plus unseen library events (ESC-50 fold 5, held-out MUSAN), with the Vaani clip itself as target, reproduces D's leaderboard SI-SDR (12.66 vs. 13.18 dB) and its ASR pattern (ΔWER $+5.1$ of a $+12.0$ ceiling, cf. leaderboard $+5.7$ vs. leaders' $\approx$17). This supports reading the task as removing only the added events while keeping native audio. We trained E13 accordingly (target = natural clip as-is, input = target + 1–3 library events; 30k mixtures from 10,206 natural clips; 6-s crops, 13,775 steps sized from a measured 5,009 steps/h).

What did not help. Detect-and-suppress gating with ground-truth timestamps on the baseline (oracle) removed speech under events; bootstrapping noise from Vaani recordings ($x - D(x)$ inside annotated events) lowered held-out scores; time-reversal TTA was worse than shift TTA at equal compute; an output high-pass filter and soft limiting reduced SI-SDR by 2–6 dB; mixing the input back (observation adding) raised ASR but cost more SI-SDR; averaging D with zero-shot MossFormerGAN, FRCRN or MossFormer2 lowered the local score; training D to remove continuous background (E12) lost 2.7 dB on held-out clips.

TABLE II
CODABENCH PHASE-2 SUBMISSIONS.

Submission	SI-SDR	ΔWER s/n (%)	Comb.
Baseline mask-net, real. sim.	–	–	3.40
MP-SENet DNS, zero-shot	–	–	10.00
MP-SENet fine-tuned (E8)	12.37	4.65 / 0.20	13.99
avg(E9-A, E9-B)	12.73	5.05 / 0.27	14.53
B+D, 4-s windows	12.86	5.41 / 0.31	14.81
D, 8-s + TTA×4	13.15	5.69 / 0.30	15.18
D, 8-s + TTA×8	**13.18**	**5.68 / 0.30**	**15.21**
D TTA×8 + peak 0.95	13.18	5.09 / 0.38	15.07
D TTA×8 + resid. gate	12.51	5.26 / 0.26	14.37

TABLE III
LOCAL EVALUATIONS (WIN8 + TTA×2 UNLESS NOTED).

Setting	D	alt.
val comb.: win4 vs win8+TTA×8	18.44	19.83
val comb.: +MossFormerGAN / +FRCRN	19.20	17.62 / 18.53
held-out SI-SDR: E12 (bg. removal)	17.25	14.57
natural+events SI-SDR: E13	15.41	15.82
12-s natural+events SI-SDR: E13	16.06	17.50
noisy Vaani+events: E13	**12.66**	**12.95**

IV. RESULTS AND CONCLUSION

Table II lists leaderboard submissions; Table III summarises local evidence ("val" = 228 held-out synthetic + 600 natural clips).

Findings. (i) Leak-free validation is essential with reused noise. (ii) The phase-aware model class and in-domain fine-tuning give most of the gain; large generic SE models are 3–5 dB behind and hurt SraVaani. (iii) Shift TTA transfers to the leaderboard (+0.37) but only about a third of its local gain; output post-processing (loudness, gating) gains locally and loses on the leaderboard, consistent with the literature that SE artefacts, not residual noise, drive ASR errors [9]. (iv) Validation and test synthetic data differ in length and in whether native noise remains in the reference; the noisy-Vaani set is the first local evaluation that reproduces our leaderboard SI-SDR.

Conclusion. A small phase-aware enhancer fine-tuned on leak-free, Vaani-like simulations with budget-sized annealing and shift TTA reaches 15.21. The diagnosed test structure suggests that event-only targets on long, noisy natural clips are the next lever; E13 improves the test-mimicking set by 0.28 dB but could not be scored on the leaderboard before the deadline.

REFERENCES

[1] Y.-X. Lu, Y. Ai, and Z.-H. Ling, "MP-SENet: A speech enhancement model with parallel denoising of magnitude and phase spectra," in *Proc. Interspeech*, 2023. Code/checkpoint: https://github.com/yxlu-0102/MP-SENet.

[2] C. K. A. Reddy *et al.*, "The INTERSPEECH 2020 Deep Noise Suppression Challenge: Datasets, subjective testing framework, and challenge results," in *Proc. Interspeech*, 2020.
[3] ARTPARK-IISc, "VAANI Noise Event Dataset: A curated spontaneous speech dataset annotated with timestamps for noise events," arXiv:2609.02474, 2026.
[4] ARTPARK-IISc, "SraVaani-1.0," https://huggingface.co/ARTPARK-IISc/SraVaani-1.0.
[5] S. Nag *et al.*, "noise-event-detection-and-removal (IndoML 2026 baseline)," https://github.com/nagshubhadip/noise-event-detection-and-removal.
[6] K. J. Piczak, "ESC: Dataset for environmental sound classification," in *Proc. ACM Multimedia*, 2015.
[7] D. Snyder, G. Chen, and D. Povey, "MUSAN: A music, speech, and noise corpus," arXiv:1510.08484, 2015.
[8] J. Le Roux, S. Wisdom, H. Erdogan, and J. R. Hershey, "SDR – half-baked or well done?" in *Proc. ICASSP*, 2019.
[9] K. Iwamoto *et al.*, "How bad are artifacts?: Analyzing the impact of speech enhancement errors on ASR," in *Proc. Interspeech*, 2022.
[10] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *Proc. ICLR*, 2019.
[11] L. N. Smith and N. Topin, "Super-convergence: Very fast training of neural networks using large learning rates," in *Proc. SPIE*, 2019.